

IT PROJECT MANAGEMENT ASSIGNMENT 2

Group Assignment

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Group 1

Question 1

Q.1.1 Any project's initiation phase requires the use of a project assessment form since it offers a formal framework for evaluating the viability, worth, and possible hazards of the suggested solution. It guarantees that choices are supported by facts rather than conjecture, bringing the project into line with stakeholder expectations and community needs. The assessment form for the Emergency SOS Services App will assist in determining whether the project can be finished in a year, will be available in low-network locations, and fits the R1.75 million budget.

Project managers can do the following by using a project evaluation form:

- Clearly define the project's goals and deliverables.
- Provide justification for the resource allocation.
- Recognize hazards early on.
- Make timelines and expenses transparent to all parties involved.
- Make sure the initiative is in line with more general community wellness and safety objectives.

Thus, the assessment form supports stakeholder buy-in and project approval by serving as a communication and decision-making tool.

Q.1.2 What is cross-platform development?

Cross-platform development is the process of creating mobile apps that work flawlessly across several operating systems (like iOS and Android) with a single codebase. This is made possible by frameworks like Flutter, React Native, and Xamarin, which offer common development environments while maintaining native-like user experiences (Brinzei, 2024; JetBrains, 2024).

How will cross-platform development contribute to meeting project timelines?

Since the team just has to maintain one codebase instead of two distinct native apps, using a cross-platform framework will speed up development. By cutting development time by up to 40%, this method allows milestones to be delivered within the project's 12-month timeline more quickly (Lehman, 2024). Additionally, it lessens the duplication of resources, which makes the budget more flexible while still satisfying functional needs.

Three potential risks of cross-platform development:

1. **Performance limitations-** Cross-platform apps, particularly those for GPS and real-time communication, could operate a little more slowly than fully native programs.
2. **Device compatibility issues-** Certain functions, such as location monitoring in the background, could act differently on iOS and Android.
3. **Framework dependency-** Long-term maintenance depends on continued support from the chosen framework's community and vendors.

How cross-platform development will enhance user experience and usability:

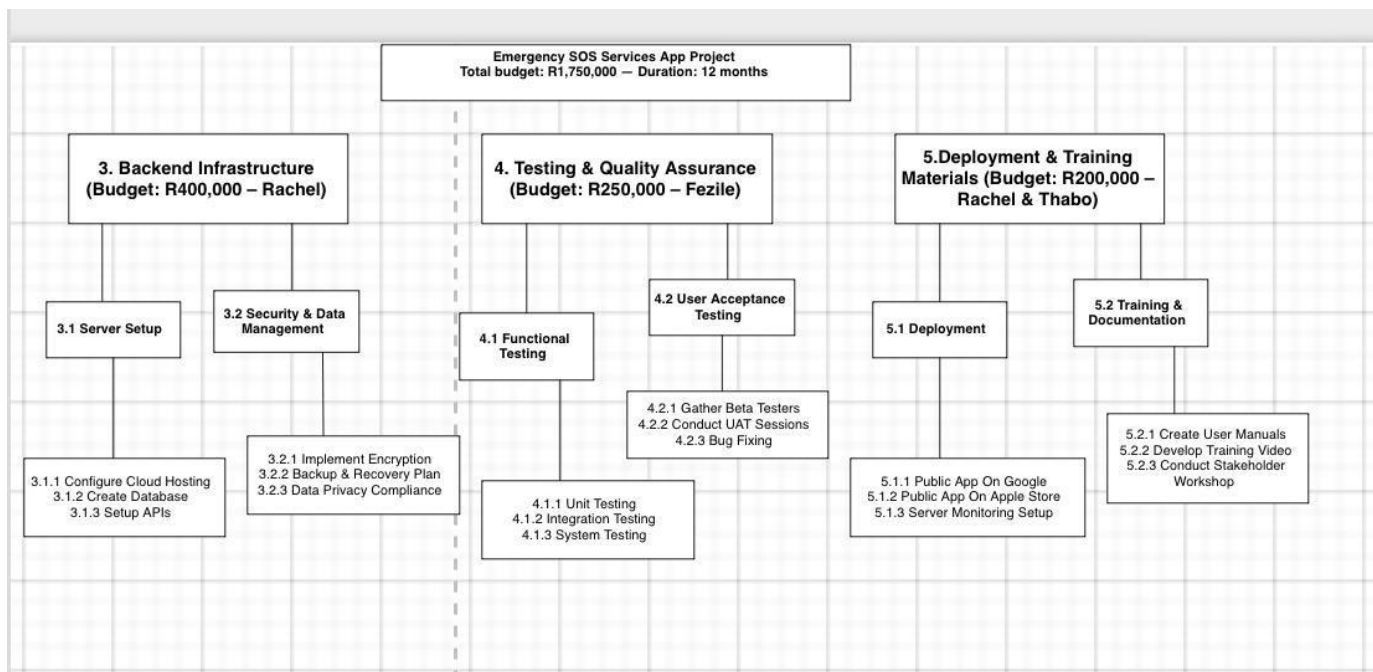
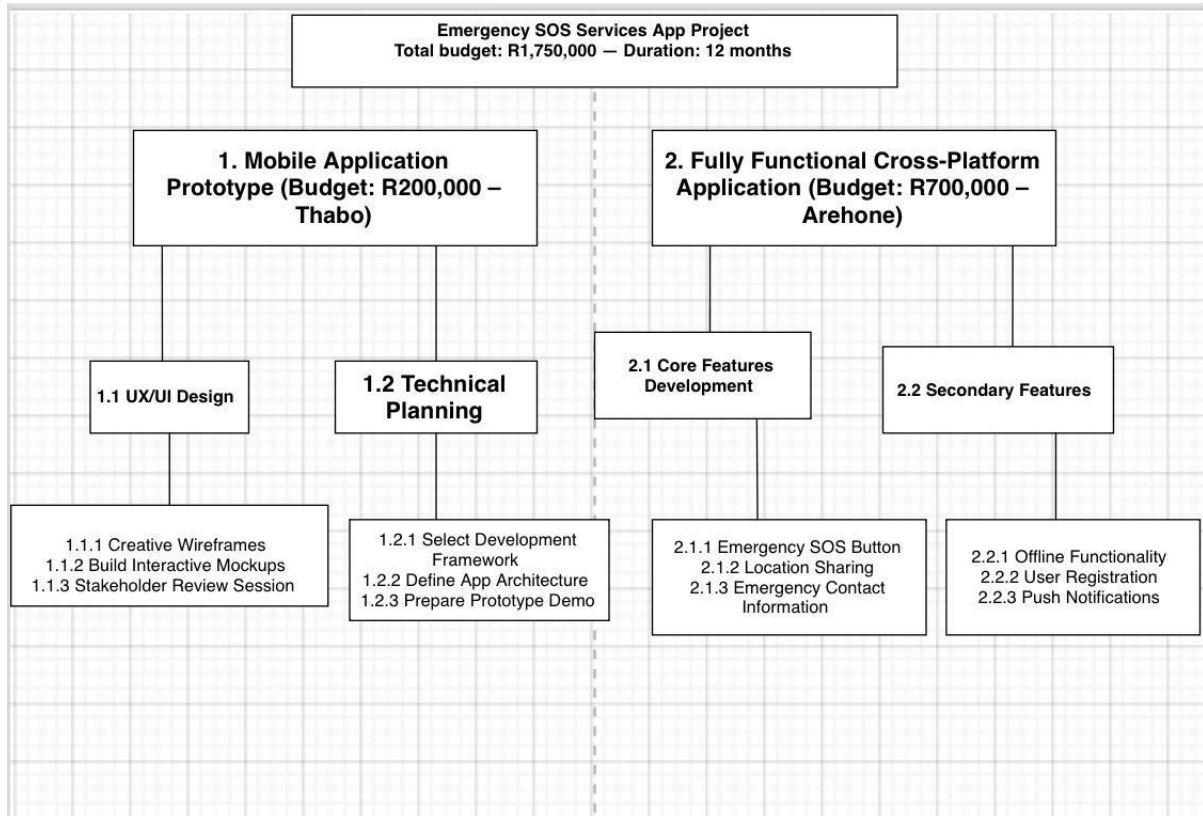
In situations when simplicity is essential, cross-platform frameworks enable consistent UI/UX across devices. Regardless of the type of device, users will encounter a consistent interface, which will minimize misunderstanding and provide prompt access to SOS functions. Furthermore, responsive designs that

preserve usability and efficiency even in low-connectivity situations are made possible by frameworks like as Flutter (Babich, 2024).

Q.1.3 Major deliverables for the Emergency SOS Services App project charter would include the following:

1. **Mobile Application Prototype-** Initial app design and wireframes for stakeholder approval.
2. **Fully Functional Cross-Platform Application-** essential functions including location sharing, emergency contact notifications, and an SOS button.
3. **Backend Infrastructure-** APIs, databases, and cloud-based servers to handle user information, requests, and communication services.
4. **Testing and Quality Assurance Reports-** Documentation for security, performance, and usability testing.
5. **Deployment and Training Materials-** published apps with stakeholder training and user manuals in the Google Play Store and Apple App Store.

Question 2



Question 3

Q.3.1

WBS	Task Name	Duration (Days)	Predecessor(s)	Resources	Cost (ZAR)
1	Application Prototype (Budget: R200,000 - Thabo)				
1.1	UX/UI Design			Thabo	
1.1.1	Create Wireframes	10		Thabo	R40,000
1.1.2	Build Interactive Mockups	12	1.1.1	Thabo	R40,000
1.1.3	Stakeholder Review Session	5	1.1.2	Thabo, Rachel	R40,000
1.2	Technical Planning			Rachel, Arehone	
1.2.1	Select Development Framework	7	1.1.1	Rachel, Arehone	R26,666.67
1.2.2	Define App Architecture	10	1.2.1	Rachel, Arehone	R26,666.67
1.2.3	Prepare Prototype Demo	5	1.2.2	Thabo, Arehone	R26,666.66
2	Fully Functional Cross-Platform Application (Budget: R700,000 - Arehone)			Arehone	
2.1	Core Features Development			Arehone	
2.1.1	Develop SOS Button	20	1.2.2	Arehone	R163,333.34
2.1.2	Develop Location Sharing	25	1.2.2	Arehone	R163,333.33
2.1.3	Develop Emergency Contact Information	15	1.2.2	Arehone	R163,333.33

WBS	Task Name	Duration (Days)	Predecessor(s)	Resources	Cost (ZAR)
15	2.1.3 Develop Emergency Contact Information	15	1.2.2	Arehone	R163,333.33
16	2.2 Secondary Features			Arehone	
17	2.2.1 Implement Offline Functionality	18	1.2.2	Arehone	R70,000
18	2.2.2 Implement User Registration/Login	12	1.2.2	Arehone	R70,000
19	2.2.3 Implement Push Notifications	10	1.2.2	Arehone	R70,000
20	3 Infrastructure (Budget: R400,000 - Rachel)			Rachel	
21	3.1 Server Setup			Rachel	
22	3.1.1 Configure Cloud Hosting	15	1.2.1	Rachel	R80,000
23	3.1.2 Create Database	12	3.1.1	Rachel	R80,000
24	3.1.3 Setup APIs	10	3.1.2	Rachel	R80,000
25	3.2 Security & Data Management			Rachel	
26	3.2.1 Implement Encryption	8	3.1.3	Rachel	R53,333.34
27	3.2.2 Backup & Recovery Plan	6	3.1.3	Rachel	R53,333.33
28	3.2.3 Data Privacy Compliance	7	3.1.3	Rachel	R53,333.33
29	4 Testing & Quality Assurance (Budget: R250,000 - Fezile)			Fezile	
					R250,000

question 3

Search for tools, help, and more (Alt + Q)

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	A	B	C	D	E	F	G	H	I	J	K	L
28	3.2.3	Data Privacy Compliance	7	3.1.3	Rachel	R53,333.33						
29	4	Testing & Quality Assurance (Budget:R250,000 - Fezile)	-	-	Fezile	R250,000						
30	4.1	Functional Testing	-	-	Fezile	-						
31	4.1.1	Unit Testing	12	2.1.1, 2.1.2, 2.1.3, 3.1.3	Fezile	R50,000						
32	4.1.2	Integration Testing	15	4.1.1	Fezile	R50,000						
33	4.1.3	System Testing	10	4.1.2	Fezile	R50,000						
34	4.2	User Acceptance Testing	-	-	Fezile,Thabo	-						
35	4.2.1	Gather Beta Testers	7	4.1.1	Fezile,Thabo	R33,333.34						
36	4.2.2	Conduct UAT Sessions	10	4.1.3, 4.2.1	Fezile, Users	R33,333.33						
37	4.2.3	Bug Fixing	12	4.2.2	Arehone,Fezile	R33,333.33						
38	5	Deployment & Training (Budget:R200,000 - Rachel & Thabo)	-	-	Rachel,Thabo	R200,000						
39	5.1	Deployment	-	-	Rachel,Thabo	-						
40	5.1.1	Publish App On Google Play	5	4.2.3	Rachel,Thabo	R33,333.34						
41	5.1.2	Publish App On Play Store	7	4.2.3	Rachel,Thabo	R33,333.33						
42	5.1.3	Setup Server Monitoring	6	3.1.3	Rachel	R33,333.33						

Workbook Statistics

question 3

Search for tools, help, and more (Alt + Q)

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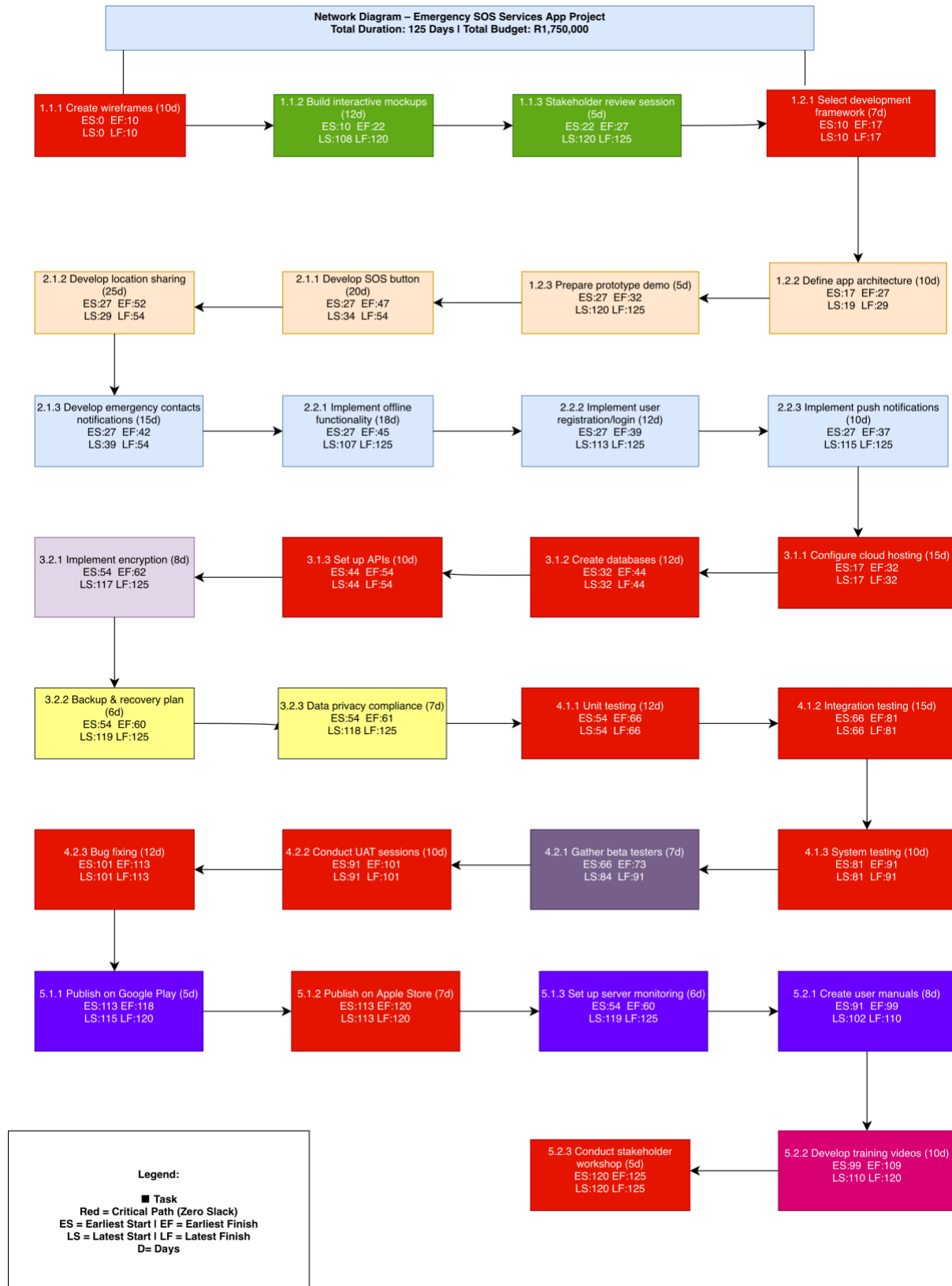
	A	B	C	D	E	F	G	H	I	J	K	L
37	4.2.3	Bug Fixing	12	4.2.2	Arehone,Fezile	R33,333.33						
38	5	Deployment & Training (Budget:R200,000 - Rachel & Thabo)	-	-	Rachel,Thabo	R200,000						
39	5.1	Deployment	-	-	Rachel,Thabo	-						
40	5.1.1	Publish App On Google Play	5	4.2.3	Rachel,Thabo	R33,333.34						
41	5.1.2	Publish App On Play Store	7	4.2.3	Rachel,Thabo	R33,333.33						
42	5.1.3	Setup Server Monitoring	6	3.1.3	Rachel	R33,333.33						
43	5.2	Training & Documentation	-	-	Rachel,Thabo	-						
44	5.2.1	Create User Manuals	8	4.1.3	Thabo, Rachel	R33,333.34						
45	5.2.2	Develop Training Videos	10	5.2.1	Thabo, Rachel	R33,333.33						
46	5.2.3	Conduct Stakeholder Support	5	5.2.2, 5.1.1, 5.1.2	Rachel,Thabo	R33,333.33						
47												
48												
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52												

Workbook Statistics

Q.3.2

Question 4

Q.4.1



Question 5

Q.5.1

Q.5.2 Introduction

Through this project, I was able to comprehend how crucial project management is to the development and implementation of a mobile community safety system. I became aware of how effective planning and communication are made possible by programs like GitHub, MS Project, and WBS.

Skills Learnt:

- **Technical Skills:** I became more proficient in creating organized documentation, monitoring issues with GitHub, and scheduling using MS Project.

- **Communication/Teamwork:** In my capacity as project manager, I assigned responsibilities, facilitated conversations, and made sure that everyone was on the same page regarding the project's objectives.
- **Management Skills:** By leading the team from inception to final submission, I improved my leadership, decision-making, and time management skills.

Role in the Team:

I oversaw and directed every stage of the project in my capacity as project leader. In addition to setting up GitHub and generating the WBS, I was also in charge of creating the MS Project schedule and writing the reflection and evaluation reports. I made the final decisions, oversaw all tasks, and made sure that deliverables were finished on schedule even when others contributed. The project remained within scope, budget, and schedule thanks to my guidance.

Research & Technology:

I personally explored cross-platform frameworks like Flutter and React Native, budget allocation tactics, and project management methodologies. Additionally, I oversaw the way information was presented using project plans, diagrams, and structured reports.

Strengths & Weaknesses:

- **Strengths:** Research, organization, communication, problem-solving, leadership, and accountability.
- **Weaknesses:** I occasionally took on excessive responsibilities, which was too much to handle. In order to complete projects in the future, I will need to get better at assigning assignments.

Stakeholder Relationships:

As team leader, I coordinated stakeholder interactions and kept my team working closely together. I made certain that all expectations were fulfilled and that problems were promptly fixed.

Impact:

The success of the project was directly impacted by my leadership. I made sure that quality requirements were fulfilled and that the project could be completed on schedule by assuming responsibility for every assignment. The team and stakeholders benefited greatly from my work.

Conclusion

All things considered, this project taught me how to use project management frameworks like Agile and PMBOK to oversee a project from start to finish. I improved my organizational abilities, developed my leadership confidence, and readied myself for future leadership positions in IT project management.

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